Mateo Rodríguez Polo

Doctoral Researcher in Stochastic Finance

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Research Profile

Stochastic control, backward stochastic differential equations, time inconsistency, filtering and hidden information, mathematical finance, market microstructure, and computational methods for stochastic optimisation.

Education

2023–present **PhD in Mathematics**, *ETH Zürich*
Stochastic Finance Group. Supervisors: Prof. Dylan Possamaï and Prof. Johannes Muhle-Karbe. Research in stochastic control, mathematical finance, hidden information, and market microstructure.

2021–2023 **MSc in Mathematics, with distinction**, *ETH Zürich*
GPA: 6.00/6.00. Willi Studer Prize; ETH-D Scholarship. Master's thesis supervised by Prof. Dylan Possamaï.

2020–2021 **Visiting Erasmus Student**, *Imperial College London*
Completed the final 60 ECTS of the bachelor's degree. GPA: 93.7% (top 1%).

2017–2021 **BSc in Mathematics**, *Autonomous University of Madrid*
GPA: 9.84/10 (top 1%). Bachelor's thesis: *Geometric Function Theory*, supervised by Prof. Dragan Vukotić; grade 10/10 with honours.

2006–2017 **Professional Degree in Music**, *Professional Conservatory of Music, Salamanca*
Specialisation: bassoon.

Professional Experience

Jun–Sep 2025 **Quantitative Research Intern**, *Jump Trading, London*
Ten-week quantitative-research internship.

Apr 2023 **Quantitative Trading Intern**, *Jane Street Capital, London*
Three-week off-cycle trading internship.

2022, 2024–2025 **Teaching Assistant**, *ETH Zürich*
Problem classes in mathematical finance, applied stochastic processes, stochastics and machine learning, numerical analysis of stochastic differential equations, and linear algebra.

2020–2021 **Grader**, *Art of Problem Solving*
Part-time grader specialising in advanced mathematical olympiad problems.

Awards and Honours

- **Willi Studer Prize** for achieving a perfect score in the ETH Zürich MSc; **ETH-D Scholarship**, awarded to 30 students across all ETH master's programmes.
- **Extraordinary Award** for the highest graduating GPA at the Autonomous University of Madrid; 200 of 240 ECTS completed with honours.
- Three **Excellence Grants from the Community of Madrid**, for the academic years 2017–2018, 2018–2019, and 2019–2020.
- **Silver medal** at the 2021 International Mathematics Competition for University Students; bronze medals in 2018, 2019, and 2020.
- **Bronze medal** at the 2017 Spanish Mathematical Olympiad.
- First place at the 2017 Baccalaureate Olympiad of the University District of Salamanca; second place in 2016.
- Top-three placements in the 2012, 2013, 2014, and 2015 Secondary Education Mathematical Olympiads of Salamanca.
- Honours in the Spanish Baccalaureate; first prize in Mathematics, Physics and Chemistry at the national Young Researchers contest.

Research Papers

[1] D. Possamaï and **M. Rodríguez Polo**. *Here, There and Everywhere: State-Dependent Time-Inconsistent Stochastic Control*. arXiv:2603.22022.

[2] F. Beretta, D. Possamaï, **M. Rodríguez Polo**, and L. Tangpi. *Policy Iteration for Exploratory Control Problems via BSDEs*. To appear.

[3] H. Pagès, D. Possamaï, and **M. Rodríguez Polo**. *Equilibrium Prices under Hidden Markov Fundamentals*. arXiv:2609.21684.

[4] J. Muhle-Karbe, R. Oomen, and **M. Rodríguez Polo**. *Information Leakage and Opportunistic Trading Around the FX Fix*. SSRN 5351658.

Conference and Seminar Talks

Jan 2024	**23rd Annual SAET Conference**. Santiago de Chile.
Jun 2024	**ETH–Hong Kong–Imperial Mathematical Finance Workshop**. London.
Jul 2024	**12th World Congress of the Bachelier Finance Society**. Rio de Janeiro.
Oct 2024	**Stochastic Control and Games for Risk and Regulation**. Hammamet.
Apr 2025	**ETH–Hong Kong–Imperial Mathematical Finance Workshop**. Hong Kong.
Apr 2025	**9th Asian Quantitative Finance Conference**. Shenzhen.
Jun 2025	**12th General AMaMeF Conference**. Verona.
Dec 2025	**IV Workshop on Stochastic Analysis and Optimization Applied to Economics, Finance, Energy and Insurance**. Puerto Natales.
Jun 2026	**Quantitative Finance Conference 2026**. Singapore.
Jun 2026	**ETH–Hong Kong–Imperial Mathematical Finance Workshop**. Zürich.
Jul 2026	**13th World Congress of the Bachelier Finance Society**. Bologna.
22 Sep 2026	**ORFE Student Seminar, Princeton University**. Princeton.

Technical Skills and Languages

Programming	Python, C, MATLAB, SageMath, Processing.
Scientific tools	LaTeX.
Languages	Spanish (native); English (C2); German (B2 Goethe-Zertifikat).

Additional Activities

Candidate Master in chess (title awarded in October 2022), with a FIDE rating of 2218. Competed in the Spanish First League and the Swiss National A League. Bassoonist in amateur orchestras. Interests include football, pádel, and tennis.